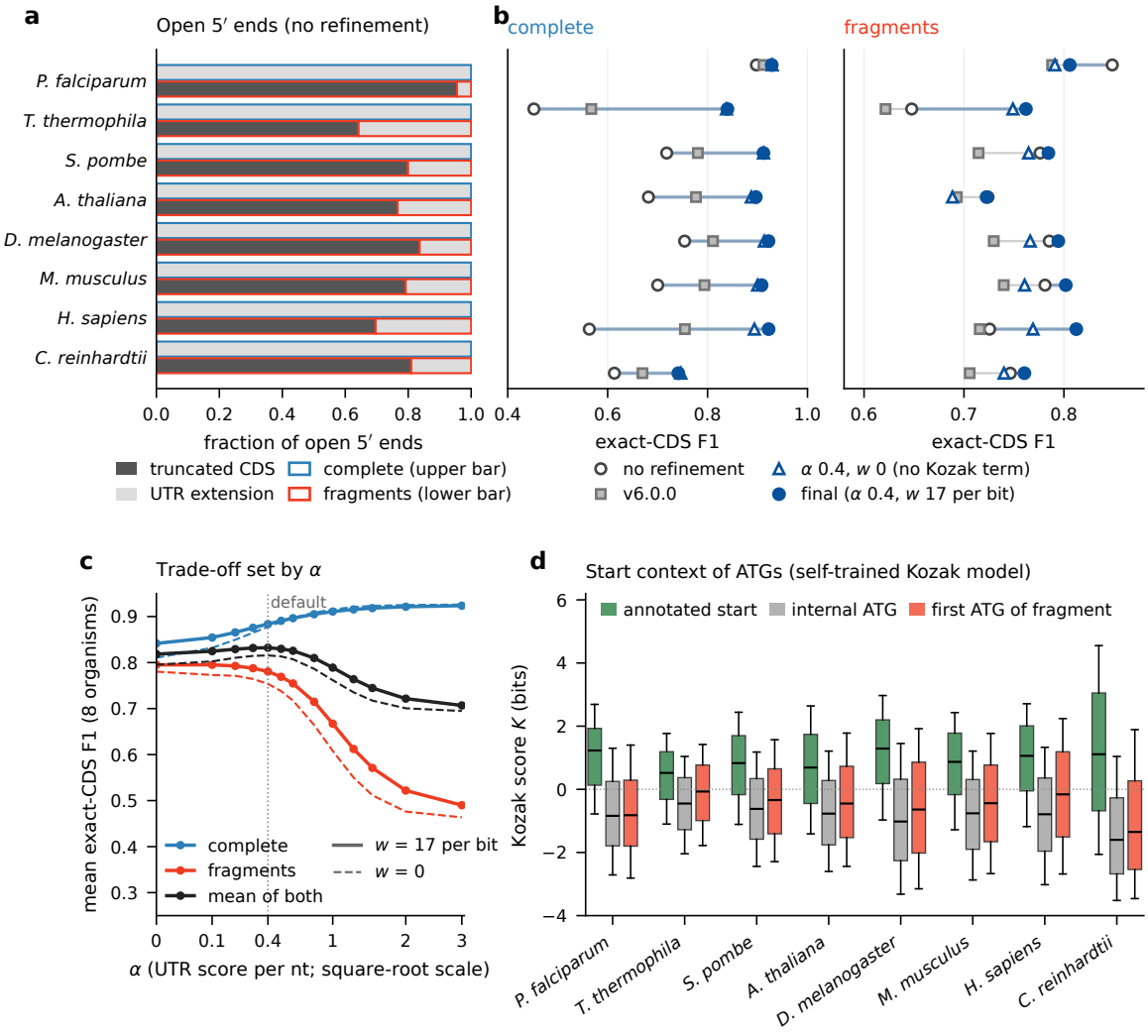
a
Open 5′ ends (no refinement)
P. falciparum
T. thermophila
S. pombe
A. thaliana
D. melanogaster
M. musculus
H. sapiens
C. reinhardtii
0.0
0.2
0.4
0.6
0.8
1.0
fraction of open 5′ ends
truncated CDS
UTR extension
complete (upper bar)
fragments (lower bar)
b
complete
fragments
0.4
0.6
0.8
1.0
exact-CDS F1
0.6
0.7
0.8
exact-CDS F1
no refinement
v6.0.0
α 0.4, w 0 (no Kozak term)
final (α 0.4, w 17 per bit)
c
Trade-off set by α
default
mean exact-CDS F1 (8 organisms)
0.9
0.8
0.7
0.6
0.5
0.4
0.3
complete
fragments
mean of both
w = 17 per bit
w = 0
0
0.1
0.4
1
2
3
α (UTR score per nt; square-root scale)
d
Start context of ATGs (self-trained Kozak model)
annotated start
internal ATG
first ATG of fragment
Kozak score K (bits)
6
4
2
0
−2
−4
P. falciparum
T. thermophila
S. pombe
A. thaliana
D. melanogaster
M. musculus
H. sapiens
C. reinhardtii